

Conversion Teardown

Why 211 paid visitors produced zero activated users — and the fixes, ranked.

Bottom line: The marketing is working — traffic is cheap and flowing. The **funnel is broken in four independent places**, each fatal on its own. No amount of ad spend converts through a locked door. **Pause paid acquisition until P0 is fixed.**

Prepared for Gabriel Gomez · Founder, Mirror · trymirror.world

Evidence-based on the live `Mirror` + `mirror-server` codebases and production data · August 2026

Executive summary

THE ONE-PARAGRAPH VERSION

A recent Instagram video ad reached 2,237 people and drove 230 link clicks at **\$0.11 per website visit** — a healthy top of funnel. Yet those ~211 visits produced **zero activated users**. The production database confirms it: every genuine human signup sits at `email_verified = 0` and `intake_completed = 0`. The only accounts that ever complete onboarding are internal `_sim` simulations, which bypass the entire user interface. The problem is not acquisition — it is a post-click funnel with multiple hard stops.

\$0.11

COST / WEBSITE VISIT

211

WEBSITE VISITS

~0

REAL SIGNUPS

0%

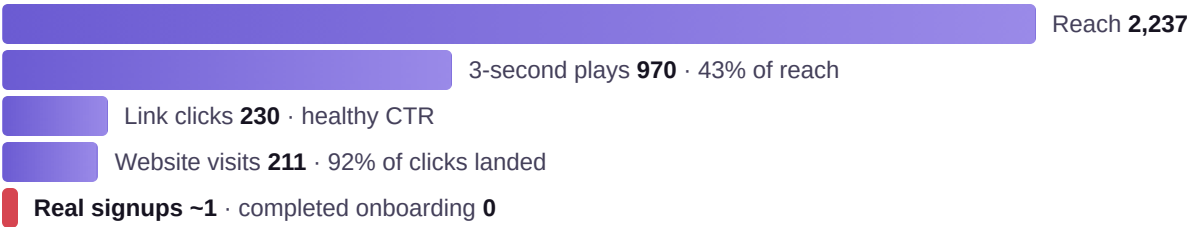
COMPLETED
ONBOARDING

FOUR FATAL WALLS, ANY ONE OF WHICH ALONE KILLS CONVERSION

- **No Meta Pixel** → Instagram optimized for cheap clicks, never signups.
- **The app's front door is a *login* screen**, not a signup — strangers have no account to enter.
- **Verification emails silently never send** — while the app tells the user one was sent.
- **Onboarding is a 25–40 minute, ~105-input gauntlet** with unskippable face + microphone walls and no payoff until the very end.

The funnel of death

WHERE ALL 211 VISITORS WERE LOST



Engagement signals were near-zero for the reach: 8 likes, 1 save, 0 follows — curiosity clicks, not intent. That is the expected shape when the platform is optimizing for clicks (no conversion signal — see Wall 0).

#	STAGE	WHAT ACTUALLY HAPPENS (FROM THE CODE)	VERDICT
0	The ad	No Meta Pixel or analytics exist anywhere in the landing page or app. With no conversion event to optimize toward, Instagram optimized for the cheapest <i>clicks</i> — producing high-volume, low-intent curiosity traffic by construction.	Built to fail
1	Landing	Long, text-dense marketing page; every CTA ("Look into the Mirror", "Start free") points at the app root, not a signup.	Leak
2	App front door	Root checks for a saved token, finds none, and routes the stranger to <code>/login</code> — a "Welcome Back / Sign In" screen for an account they don't have. Signup is only a small footer link.	Fatal
3	First paint	The login screen ships TensorFlow + face-api + three.js in the main bundle (no code-splitting) and runs a live WebGL animation — a multi-megabyte download on mobile data just to reach the login form.	Bounce
4	Signup form	Five gated fields: strict 5-rule password, live username availability, breach (HIBP) check — heavy for a \$0.11 curiosity click.	Leak
5	Verify email	The verification email is silently never sent (provider unconfigured and/or sender-domain mismatch), while registration hardcodes "email sent: true" and tells the user one went out. <code>email_verified</code> stays 0 forever.	Fatal
6	Intake	Immediate "upload your face" wall with no skip , then a microphone-recording wall with no skip , then ~105 questions (IQ + astrology + 75-question personality). 25–40 minutes, zero payoff until the final screen.	Fatal
7	Fragility	Photo and voice are held only in browser memory and wiped on any refresh or app-switch — so a mobile user interrupted mid-intake loses everything and is bounced back to re-capture.	Fatal

WHAT THE DATA PROVES

Filtering out internal test rows, keyboard-mash junk, and `__sim__` accounts, the production users table holds ~3 plausibly-real outside users ever — **none** with `intake_completed = 1`. Every `intake_completed = 1` is a simulation or internal test account. Recent ad-era signups sit at `email_verified = 0`. Real-user activation is literally 0%.

Evidence · Acquisition & entry

WALLS 0-4 · [CLICK](#) → [LOGIN](#) → [SIGNUP](#)

Wall 0 — No conversion tracking

No Google Analytics, no Meta/Facebook Pixel, no Plausible/PostHog/Mixpanel — verified across [landing/index.html](#), [client/index.html](#), and all of [client/src](#). Consequences: (1) Instagram had no signup event to optimize toward, so it bought clicks; (2) you cannot see *where* the 211 drop off. You are flying blind and buying the wrong thing.

Wall 2 — The front door is a login screen

The app root is gated by [IntakeGate](#). With no stored token it runs [navigate\('/login'\)](#) (App.tsx:82), rendering "Welcome Back / Sign In" (Login.tsx:504,587). Signup is a footer link only (Login.tsx:898–907). An ad inviting a stranger to "discover yourself" lands them on a returning-user login form — the single biggest click-to-signup killer.

Wall 3 — Multi-megabyte first paint

No route-level code-splitting: [App.tsx](#) statically imports every page. Because it eagerly pulls [IntakeFlow](#) → [VisualStep](#) → [useFaceApi](#) → [@vlamandic/face-api](#), TensorFlow ([@tensorflow/tfjs](#)) + the WebGL backend land in the main entry chunk, alongside [three](#), [@react-three/fiber](#), framer-motion, and [html2pdf](#). The login/register screens also render a live three.js scene (Login.tsx:63, RegistrationStep.tsx:39). Result: a heavy download and continuous WebGL animation on low-end mobile before the login form is even usable.

Wall 4 — Signup effort vs. intent

Five progressively-disclosed required fields, a password requiring upper + lower + number + symbol at 8+ chars, live username availability, and an online breach check (RegistrationStep.tsx:393–415, 558–565, 104–128). Reasonable for a committed user; far too much for a curiosity click.

Evidence · The verification email (Wall 5)

THE HIGHEST-CONFIDENCE, HIGHEST-IMPACT BUG

Exactly one code path flips `email_verified = 1`: the user clicking the emailed link and their browser POSTing the token back (`emailVerificationController.ts:247`). If the email never arrives, no user is ever verified — and the client gates intake on that flag.

THE CHAIN OF SILENT FAILURE

- The email service **disables itself** when neither `EMAIL_API_KEY` nor `RESEND_API_KEY` is set — logging only a warning; every send becomes a no-op (`emailService.ts:468–475, 522–525`).
- The initial verification email is **queued** to Redis and returns success without checking whether the provider is enabled; the queue processor then **early-returns when disabled**, so the item is never sent and the token expires in 24h (`emailService.ts:583–625, 632`).
- Registration nonetheless returns a **hardcoded** `verification.emailSent: true` and "We just sent you a verification email" — the UI reports success regardless (`authController.ts:578`).
- Provider errors (e.g. a 403 "domain not verified") are caught and only logged; the queue retries 3× then drops the item permanently and silently (`emailService.ts:363–372, 656–674`).

COMPOUNDING: THE SENDER-DOMAIN MISMATCH

Every email defaults to `From: noreply@theundergroundrailroad.world` with verify-links to `APP_URL = theundergroundrailroad.world/Mirror` (`emailService.ts:347,398,461,463`), while the live app/marketing origin is `trymirror.world`. If the provider's *verified* sender is one domain and mail is sent from the other, sends are rejected (403) — silently. And if users live on `trymirror.world`, verify-links may point at the wrong host. **This is the same deliverability issue from the mailbox work — now product-critical.**

FIRST THING TO CHECK (5 MINUTES, OPERATIONAL)

On the production host: is `RESEND_API_KEY` / `EMAIL_API_KEY` set and non-empty, `EMAIL_PROVIDER` correct, and `EMAIL_DRY_RUN` not `true`? Hit `GET /mirror/api/health` — it reports the email service as `disabled` / `unhealthy` directly. Strong hypothesis: it is disabled, and that one env var is why the funnel is dark.

Note: the `users.public_key` / `private_key` / `key_generated_at` columns are NULL for everyone including sims — dead, unused schema. Not part of the problem; not to be conflated with the email bug.

Evidence · The onboarding gauntlet (Wall 6–7)

WHY EVEN VERIFIED USERS DON'T FINISH

"Completing intake" means one API call succeeding — `POST /intake/store` (`intakeController.ts:719`) — the *last* step of a 7-stage UI funnel. A cold user must survive all of it before seeing anything.

#	STEP	WHAT THE USER MUST DO	HARD STOP?
1	Registration	Username + email + strict password + terms; then a forced 3.5s wait before advancing.	Validation
2	Visual	Take/upload a photo and pass on-device face detection . Code comment: "there is intentionally no skip path."	YES
3	Vocal	Grant microphone permission and record ≥ 3 seconds. No mic → cannot proceed.	YES
4	IQ	Answer 30 questions .	Required
5	Astrology	Manually enter birth date + exact time + place.	Required
6	Personality	~ 75 questions (advertised "12–18 minutes"), including free-text reflections needing ≥ 20 characters.	Required
7	Submit	Three sequential network calls (photo upload, voice upload, store). Any failure on weak mobile data drops the user at the finish line.	YES

~105+ INPUTS · PHOTO · VOICE · 25–40 MINUTES · ZERO PAYOFF UNTIL THE LAST SCREEN

Results — the first and only glimpse of "their Mirror" — is gated behind `intakeCompleted === true` (`App.tsx:316,332,362`), which only flips after step 7 succeeds. There is no teaser, no partial reveal, no early "aha."

Why simulations pass and humans don't

The simulation harness never touches the UI. It calls the same backend endpoints directly over loopback with synthetic data — a 1×1 PNG and a fabricated face-analysis object instead of real face detection, a silent WAV instead of a microphone, hardcoded IQ/astrology/personality fixtures (`intakeSimulationController.ts:771,815,824,879`). The backend pipeline is healthy; that is precisely why sims reach `intake_completed = 1`. Real humans fail because reaching the same endpoint requires surviving the entire client gauntlet.

Fragility that punishes mobile users

- **Media lost on interruption:** `IntakeContext` strips `photo` and `voice` before writing to `localStorage` (`IntakeContext.tsx:117`); they live only in memory. A refresh, an incoming call, or app-switching during the 25–40 min flow loses them, and Submit then bounces the user back to re-capture.
- **Route-order trap:** bare `/intake` redirects to Personality (step 6), not the real start at Visual (`IntakeFlow.tsx:17`); an interrupted user can land mid-flow with empty context and fail Submit.

The fix list, ranked by impact-per-effort

P0 — This week · small changes that unblock everything

- 1 Pause paid ads.** Stop paying \$0.11 for clicks that cannot convert. Resume once P0 is deployed. effort: zero
- 2 Fix the verification email.** Set the provider key on prod; reconcile `EMAIL_FROM_ADDRESS` + `APP_URL` with the provider's verified sending domain; confirm via `/health`. Nothing converts until this works. effort: low
- 3 Send the logged-out front door to `/register`, not `/login`** (`App.tsx:82`). Near one-line change; enormous return. effort: low
- 4 Install a Meta Pixel + lightweight analytics** on landing and app, firing a signup event — so Instagram optimizes for customers and you can see the drop-off. effort: low

P1 — Next · remove the hard walls

- 1 Make the face + voice steps skippable** (remove "no skip path"), or move them deep into the intake instead of first. effort: med
- 2 Deliver value before the 40-minute ask** — a short teaser result after a few questions, then invite the deeper intake. effort: med
- 3 Stop the "email sent" lie** (`authController.ts:578`) — surface real send status so failures are visible to you and the user. effort: low

P2 — Then · harden

- 1 Code-split** the app (lazy-load intake, face-api, TensorFlow, three.js) so login/register isn't gated behind a multi-MB download. effort: med
- 2 Persist photo/voice** to IndexedDB so a refresh or app-switch doesn't destroy the session (`IntakeContext.tsx:117`). effort: med
- 3 Trim friction:** stage/shorten the intake, ease the forced 3.5s wait, fix the `/intake` → Personality misroute, and consider a dedicated ad landing page with one signup CTA above the fold. effort: low

RECOMMENDED FIRST MOVE

The two highest-return, lowest-effort changes are **#2 (verification email)** and **#3 (front door → signup)**. Together they turn a dead funnel into a working one. Add **#4 (Pixel)** so the next ad dollar is spent on conversions, not clicks — then, and only then, turn acquisition back on.